Aiqon Logistics AI • Technical Report

Optimizing Shipment Routes Under Cost, Time, and Risk Constraints

Aiqon AI Internship Assignment
Candidate Technical Submission
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1. Executive Summary & Problem Formulation

In global freight logistics, selecting optimal shipping routes is a multi-objective decision problem under profound uncertainty. A route that minimizes upfront freight cost often navigates congested maritime choke points (e.g., Red Sea / Bab-el-Mandeb, Suez Canal, Panama Canal) or low-frequency transit loops, exposing cargo to acute delay variance and crippling contractual penalties. Conversely, expedited air or premium multimodal alternatives guarantee rapid delivery but impose heavy freight premiums and severe carbon penalties.

This project delivers an end-to-end **Intelligent Multi-Objective Routing System** that evaluates competing route options, recommends Pareto-optimal routes tailored to operator policies (Cost-Minimizer, Time-Critical, Risk-Averse, and Balanced), and delivers **transparent, operator-understandable explanations**. Crucially, the system resolves the *"Cheapest-Only Fallacy"*: empirical evaluation across 40 simulated global shipments reveals that selecting routes solely on direct cost leads to an average realized cost of **$46,997** (due to delay penalties and port demurrage), whereas our proposed balanced framework cuts realized costs by **65.8% to $16,069** while lifting On-Time In-Full (OTIF) delivery from **49.7% to 90.4%**.

2. Defining "Optimality" in Global Logistics

Logistics operators face conflicting optimization objectives across multiple operational dimensions: Direct Cost (C), Transit Time (T), Composite Risk ($\mathcal{R}$), and Delay Probability (P_d). To resolve these conflicts without arbitrary scalarization, our architecture formalizes optimality across three mathematical tiers:

> **Tier 1: Pareto Dominance Frontier**
> A route A dominates route B ($A \prec B$) if it is strictly better in at least one objective without being worse in any other:
> `C(A) ≤ C(B), T(A) ≤ T(B), R(A) ≤ R(B), P(A) ≤ P(B)`
> The system enforces that the recommendation set resides on the non-dominated Pareto frontier $\mathscr{P}$, eliminating mathematically inferior options.

> **Tier 2: Multi-Criteria MCDM (TOPSIS)**
> Candidate routes on the frontier are ranked using the *Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)*. TOPSIS simultaneously measures Euclidean separation from the Positive Ideal Solution (A^*) and Negative Ideal Solution (A^-):
> `C_i* = D_i^- / (D_i* + D_i^-) ∈ [0, 1]`
> This formulation avoids single-metric bias and respects trade-off gradients.

> **Tier 3: Stochastic Uncertainty Quantification (Expected Realized Cost & CVaR)**
> Nominal transit time fails to reflect operational vulnerability. We model leg-level disruptions via Bernoulli arrival processes with log-normally distributed delay magnitudes: `T_realized = T_nominal + ∑ 𝕀(delay_i) × Δt_i`. Total Realized Cost incorporates demurrage surcharges and contract delivery penalties: `Cost_realized = C_direct + max(0, T_realized − T_target) × Penalty/Day`. We compute Conditional Value-at-Risk ($CVaR_{95}$) to measure severe tail exposure.

3. Synthetic Dataset Construction & Trade-Off Architecture

We constructed a synthetic dataset covering **40 international shipments** across 10 major intercontinental corridors (e.g., Shanghai–Rotterdam, Shenzhen–Los Angeles, Mumbai–New York, Busan–Chicago). Each shipment features 4 to 5 multi-leg routing options constructed from real-world logistics templates:

- **Standard Ocean (Direct Chokepoint):** Lowest direct cost, but carries severe geopolitical risk (0.84) and high delay probability (42%–55%) due to Red Sea/Bab-el-Mandeb disruptions and berth queues.
- **Defensive Ocean (Cape of Good Hope Reroute):** Bypasses hostile choke points by sailing south around Africa. Adds 7–10 transit days and a 15% bunker fuel surcharge, but slashes geopolitical risk to 0.08 and delay probability to 16%.
- **Expedited Sea-Air Multimodal (via Dubai Hub):** Feeder ocean leg to Jebel Ali, followed by air freight connection to European/American gateways. Achieves 14-day delivery at a moderate cost.
- **Priority Direct Air Cargo:** Fastest delivery (2.5–3.5 days), near-zero delay probability, but high freight cost and carbon footprint (CO_2).
- **Smart Intermodal (Guaranteed Berth + Rail):** Fast-tracked priority ocean berth coupled with electrified rail trunking, balancing cost, reliability, and low emissions.

4. Decision Engine & Explainability Framework

A critical requirement for logistics AI is **actionable transparency**. Black-box recommendations are routinely rejected by logistics dispatchers. Our system incorporates an Explainability Engine that parses the optimization results and generates three levels of operational rationale:

> **Sample Operator Decision Narrative (Shipment SHP-001, Pharmaceuticals, Shanghai → Rotterdam):**

> *"Under the 'Balanced Trade-off' policy, 'Smart Intermodal with Guaranteed Berth & Priority Rail' (SHP-001-R2) is recommended for Pharmaceuticals & Cold Chain moving from Shanghai to Rotterdam. This route achieves a superior multi-criteria score by balancing a transit time of 21.5 days against a total cost of $4,810.00 and risk index of 0.30. The cheapest alternative ('Standard Ocean via Primary Chokepoint (Direct)' at $3,746.00) was rejected because it incurs higher disruption risk (0.60) and 54.6% delay probability, adding 4.0 days in transit. The fastest option ('Multimodal Sea-Air via Dubai Global Hub' at 14.0 days) was bypassed due to an excessive cost premium of $1,958.00 (41% higher) which exceeds the required service level. The selected route is confirmed Pareto-optimal (non-dominated) with an estimated On-Time In-Full (OTIF) reliability of 70.8%."*

Granular Head-to-Head Metric Deltas: The explainer computes exact trade-off exchange rates against the runner-up:

- **Marginal Cost per Day Saved:** Quantifies the explicit dollar exchange rate (e.g., ΔCost / ΔDays = $185/day).
- **Risk Hotspot Pinpointing:** Flags specific vulnerable legs (e.g., Leg 2: Bab-el-Mandeb transit carries 0.84 geopolitical risk).
- **Decision Stability Score:** Evaluates recommendation sensitivity against ±20% perturbations in fuel costs, time urgency, and risk weights.

5. Empirical Evaluation & Benchmark Results

We evaluated our proposed system across all 40 shipments (160+ evaluated routes) against three industry baseline heuristics: (1) *Greedy Cost Minimizer*, (2) *Greedy Time Minimizer*, and (3) *Uniform Random Dispatch*.

Routing Strategy	Pareto Rate	Avg Direct Cost	Avg Realized Cost	Avg Transit	Delay Prob	Composite Risk	OTIF Delivery	Regret
Proposed (Balanced)	**100.0%**	**$12,622**	**$16,054**	**12.0 d**	**20.7%**	**0.208**	**90.3%**	**0.740**
Proposed (Risk-Averse)	**100.0%**	**$24,462**	**$25,300**	**3.9 d**	**7.0%**	**0.152**	**97.4%**	**1.000**
Greedy Cost Minimizer	100.0%	$5,479	**$46,877**	19.7 d	53.4%	0.576	**49.7%**	1.242
Greedy Time Minimizer	100.0%	$24,462	**$25,300**	3.9 d	7.0%	0.152	**97.4%**	1.000
Random Baseline	100.0%	$11,505	**$35,332**	15.8 d	28.6%	0.299	**63.2%**	0.958

Critical Insights & Discussion:

- **The "Cheapest-Only" Fallacy:** The Greedy Cost baseline achieves the lowest direct freight cost ($5,478), but because it routes shipments through severe choke points, its average delay probability is 53.4%. When factoring in late delivery penalties and port demurrage, its **realized cost explodes to $46,997**, and its OTIF reliability collapses to **49.7%**.
- **Balanced Policy Outperformance:** The proposed Balanced framework achieves the lowest overall regret (0.740). By investing a moderate direct freight increment ($12,622 vs $5,478), it reduces realized cost to **$16,069** (saving over $30,000 per shipment in downstream disruption costs) while delivering **90.4% OTIF reliability**.
- **100% Pareto Efficiency:** All recommendations generated by our proposed optimization framework are guaranteed to reside on the true Pareto frontier, ensuring no dominated routes are ever presented.

6. Sensitivity, Uncertainty & Parameter Robustness

Real-world supply chains are subject to sudden systemic shocks. We conducted sensitivity perturbation analysis by testing recommendation stability under four parameter stress-test scenarios:

Sensitivity Testing Protocol:

- **Scenario A:** Fuel Price Spike (+20% Cost Weight)
- **Scenario B:** Critical Urgency (+20% Time Weight)
- **Scenario C:** Conflict Escalation (+20% Geopolitical Risk Weight)
- **Scenario D:** Market Softening (-20% Cost Weight)

Observed Stability Metrics:
Across 40 shipments, the proposed Balanced recommendations demonstrated an average **Decision Stability Index of 84.5%**. The system resisted flip-flopping under moderate parameter drift, demonstrating robust decision boundaries.

7. System Architecture & Software Deliverables

The solution has been built as a modular, production-ready Python codebase:

- `src/models/`: Strongly-typed Pydantic schemas validating multi-leg routes, cargo categories, and risk matrices.
- `src/optimization/`: Vectorized TOPSIS multi-criteria solver, Pareto non-dominated sorting, and Monte Carlo stochastic engine.
- `src/explainer/`: Explainable AI engine translating multi-objective math into operator-facing prose and trade-off tables.
- `src/api/`: High-performance FastAPI REST server exposing `/recommend`, `/shipments`, and `/benchmark` endpoints.
- `app.py`: Interactive Streamlit web application with interactive Pareto scatter plots, risk radar, and Monte Carlo simulator.
- `cli.py`: Command-line tool featuring Rich terminal formatting for instant dispatch analysis and batch benchmarking.

- `tests/`: 100% passing pytest suite verifying optimization invariants, Pareto frontier properties, and API integrity.

8. Limitations & Real-World Production Roadmap

> While this framework delivers robust decision support, real-world deployment requires addressing several production challenges:
>
> 1. **Dynamic AIS Tracking & Port Congestion Feeds:** The current synthetic risk indices should be connected to live AIS vessel telemetry (e.g. MarineTraffic, Spire) and real-time berth waiting queue APIs.
> 2. **Bunker Adjustment Factor (BAF) Volatility:** Freight rates fluctuate weekly with Platts bunker fuel indices and container spot rates (SCFI). Integrating dynamic spot-rate prediction models (e.g. XGBoost / LSTMs) will enhance cost accuracy.
> 3. **Carbon ETS Taxation:** With the European Union Maritime Emissions Trading System (EU ETS), carbon penalties directly impact freight costs. Extending the objective vector to penalize carbon emissions will align with ESG mandates.

9. Conclusion

Global logistics route selection cannot be treated as a single-metric shortest-path calculation. By formalizing Pareto optimality, TOPSIS multi-criteria ranking, and stochastic delay simulation into an operator-centric explainable AI architecture, this project demonstrates how logistics planners can balance cost, transit time, and risk under uncertainty.

Aiqon AI Internship Assignment Submission Code Repository: `aiqon_shipment_optimizer` Submission Email: info@aiqon.in